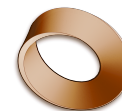


Name: \_\_\_\_\_

Date: \_\_\_\_\_

Mr. Rawson • Y9 MYP Physics



## 1.3 – Period of a Pendulum Investigation

*Unit 1: Oscillations & Waves*

*Criterion B — Inquiring and designing • Criterion C — Processing and evaluating*

### Start here, and go slowly.

Criterion B is not a list of things a teacher marks at the end. Read it in order and it is **the investigation itself**: first you say what you are trying to find out, then what you think will happen and why, then what you will change and what you will keep the same, and only then how you will actually do it.

**That order is not an accident.** You cannot write a sensible method until you know what you are controlling, and you cannot know what to control until you have a question. Work down the criterion and it walks you through planning an experiment — which is why we do this part *before* we touch a pendulum.

### Part 1 — Your investigation

Everything from here is **Criterion B — inquiring and designing**. Keep the **Criteria B and C** sheet open beside this one: each question below is labelled with the strand it is marked against.

**Every group is testing something different.** You all use the same pendulum and the same timing method. What changes is **the one thing you are allowed to change** — your **independent variable**. Write yours in the box below before you answer anything else.

| Grp | Your variable                       | What you change, and how                                                                                                                                               |
|-----|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <b>Length</b> of the string         | Shortest to longest. Measure from the pivot to the <i>centre</i> of the bob, every time.                                                                               |
| 2   | <b>Mass</b> of the bob              | Same string length, same size of bob if you can — swap the mass only.                                                                                                  |
| 3   | <b>Size</b> of the bob              | Same mass, different diameter. Harder to arrange: ask before you start.                                                                                                |
| 4   | <b>Release angle</b> (displacement) | Same length, same bob. Release from a small angle, then wider. Use a protractor — “about the same” is not a measurement.                                               |
| 5   | <b>Gravity</b>                      | <b>You cannot change gravity in this room.</b> Use the <b>PhET Pendulum Lab</b> simulation, which lets you set it — Earth, the Moon, Jupiter, and a slider in between. |

Our group is number \_\_\_\_\_ our independent variable is \_\_\_\_\_

### Read this before you collect a single number.

**“It made no difference” is a real result, and it is worth full marks.** Some of the variables on that list genuinely change the period. Some genuinely do not. Nobody in the room knows which is which until the data is in — that is the entire point of testing.

A flat graph is a *finding*. It is not a failed experiment, and it is not a sign you did it wrong. **The one thing that will cost you marks is changing your numbers to make a line appear.**

Name: \_\_\_\_\_

Date: \_\_\_\_\_

### B.i What are we trying to find out?

1) **Describe** the problem you are testing. Write it as a question, with your own variable in it — then say, in a sentence, why anyone would want to know the answer. B.i

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### B.ii What do you think will happen?

2) **Outline and explain** your hypothesis. Say what you think your variable will do to the period **and why** — a prediction with no reasoning behind it caps you at the bottom bands. B.ii

#### Sentence Starters

Fill this in and you have written a hypothesis:

As the \_\_\_\_\_ increases, the period will \_\_\_\_\_,  
because \_\_\_\_\_.

The first blank is the variable your group is changing. The second is one of *increase*, *decrease* or *stay the same* — all three are real answers.

**Check.** The word *because* has a reason written after it. That reason is where the extra marks live: a prediction scores the bottom band on its own, however well it turns out to match the data.

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*Write it down now, and then leave it alone. A prediction you quietly edit once the results are in front of you is not a prediction — and being wrong costs you nothing here.*

### B.iii Variables, and the data you will collect

#### Sentence Starters

**Resolution** is the smallest change an instrument can show. Name it whenever you name a measuring tool: “a metre ruler with a resolution of 1 mm”, “a stopwatch with a resolution of 0.01 s”.

A **control variable** needs three things written about it, in this order:

- |   |                 |                                                                                                             |
|---|-----------------|-------------------------------------------------------------------------------------------------------------|
| 1 | what it is      | <i>“I will keep the mass of the bob the same.”</i>                                                          |
| 2 | how you hold it | <i>“I will use the same bob for every measurement.”</i>                                                     |
| 3 | why it matters  | <i>“A heavier bob might swing differently, so changing it would mean I was testing two things at once.”</i> |

**Check.** Every control variable on your list has all three parts. A list that names the variables and stops is the most common way to sit a band lower than the work deserves.

3) **State** your **independent** variable (the one you change) and your **dependent** variable (the one you measure). B.iii

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4) **Describe** how you will change it — *how many* different values, and spread how far apart? B.iii

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5) **State** your **control** variables — everything you must keep the same for the test to be fair. Include the things that are easy to forget. B.iii

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6) **Describe** how you will measure the period — with what instrument, how many times, and how you will record it. *Write your best answer now; you will come back and improve it after Part 4.* B.iii

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**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

### B.iv Your method

7) **Design** your method. Number every step. List your equipment, and include one safety point.

*B.iv*

*Test it before you hand it in: give it to another group and see whether they could run it without asking you a single question. That is exactly what “logical, complete and safe” means on the criteria sheet.*

[illegible]

**What happens next.** You run the method you just wrote, and the numbers you get become **Criterion C — Part 2 of this sheet**. Your table, your graph, what it shows, and whether it agreed with the hypothesis you wrote before you started.

Then every group reports back. Between us we will have tested five different variables on the same pendulum, and **the pattern across all five groups is the actual physics** — more than any one group could find alone.

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Part 2 — Your results

Everything from here is **Criterion C — processing and evaluating**. Same rule as before: each question is labelled with the strand it is marked against, so keep the **Criteria B and C** sheet beside you.

**C is the half most people lose marks on, and it is the easier half.** B asks you to design something. C only asks you to **report honestly what happened** — but it has to be presented so that a stranger could read it without you standing there explaining.

### C.i Presenting your data — the table

Here is what a results table should look like. This one is from the **spring experiment** you have already done.

| Mass<br>/ g | Force<br>/ N | Extension<br>trial 1 / cm | Extension<br>trial 2 / cm | Extension<br>trial 3 / cm | Mean extension<br>/ cm |
|-------------|--------------|---------------------------|---------------------------|---------------------------|------------------------|
| 0           | 0.00         | 0.0                       | 0.0                       | 0.0                       | 0.0                    |
| 100         | 0.98         | 3.7                       | 4.1                       | 3.9                       | 3.9                    |
| 200         | 1.96         | 8.0                       | 7.8                       | 8.1                       | 8.0                    |
| 300         | 2.94         | 11.5                      | 11.7                      | 11.6                      | 11.6                   |
| 400         | 3.92         | 15.9                      | 15.7                      | 16.0                      | 15.9                   |
| 500         | 4.90         | 19.4                      | 19.6                      | 19.5                      | 19.5                   |

#### Required elements for a data table

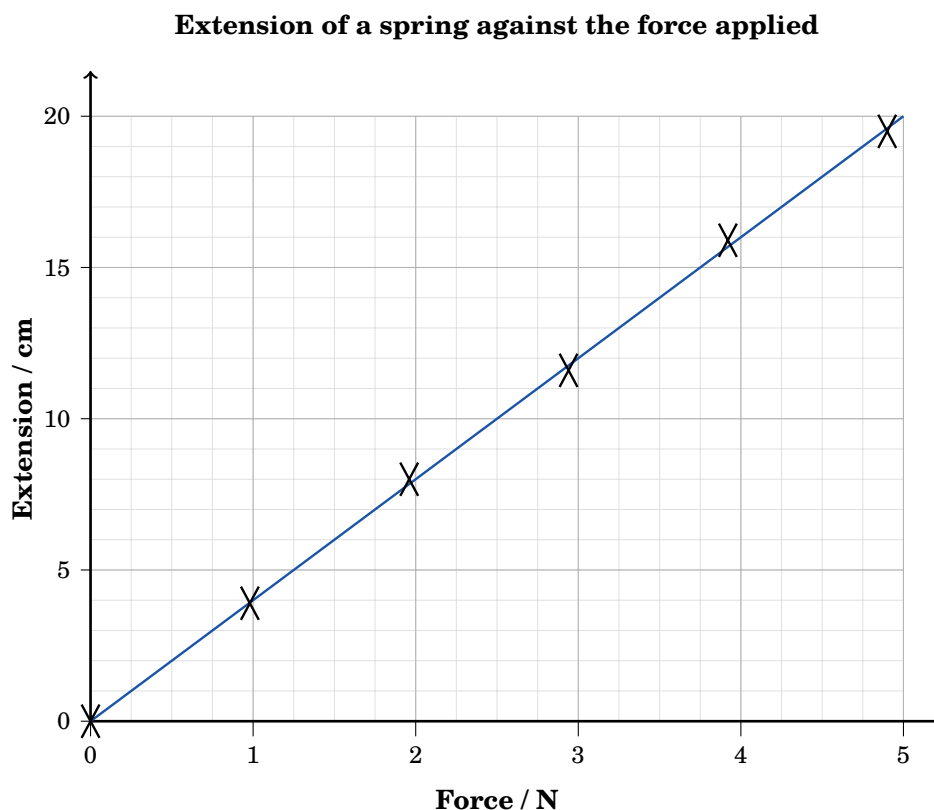
- **Headings** — with the unit, written once, in the heading.
- **Multiple trials** — three of them, then a mean.
- **Same decimal places** — all the way down a column.
- **A zero row** — nothing added, nothing measured.
- **Independent variable on the left**, dependent variable on the right.
- **Ruled lines** — round every cell.

1) **Draw** your own results table below, for **your** variable. Head every column, put the unit in the heading, and leave room for three trials and a mean. C.i

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**C.i Presenting your data — the graph**

And here is the same spring data, drawn as a graph.

**Why that graph would score well.**

- **It has a title** saying what is plotted against what.
- **Both axes are labelled, with units** — *Force / N*, *Extension / cm*.
- **The thing you changed is along the bottom**; what you measured goes up the side.
- **The scale uses most of the paper** and goes up in sensible steps. Squeezing the whole graph into one corner throws away accuracy.
- **Points marked with a small ×**, not fat dots you cannot read a value from.
- **One straight line of best fit, drawn with a ruler** — through the trend, with points either side of it. **Never dot-to-dot.**

2) **Draw** this **same** graph yourself, on a piece of graph paper, using the numbers from the table on the previous page. Then do the same for **your own** results. C.i

*Six crosses and a ruled line. Every item on the green list above has to be on your version — that list is the mark scheme.*

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**C.ii What does your data show?****Sentence Starters**

| Your graph looks like                         | So you write                                                                                                      |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Rising                                        | As $x$ increases, $y$ increases.                                                                                  |
| A straight line <b>through</b> the origin     | $x$ is directly proportional to $y$ , because the graph is a straight line through the origin.                    |
| A straight line that <b>misses</b> the origin | There is a linear relationship between $x$ and $y$ , because the graph is a straight line that misses the origin. |
| Curving upwards                               | As $x$ increases, $y$ increases at an increasing rate.                                                            |
| Rising, then levelling off                    | As $x$ increases, $y$ increases — but after $x = \rule{1cm}{0.4pt}$ , $y$ stops increasing.                       |
| Flat                                          | There is no relationship between $x$ and $y$ : changing $x$ does not change $y$ .                                 |

Swap  $x$  and  $y$  for your own two variables, then say **why** in one of three ways:

- ...because when the length was 20 cm the period was 0.9 s, and at 40 cm it was 1.3 s — numbers off your own graph;
- ...because the graph has a positive gradient;
- ...because the graph is curving upwards.

**Check.** Your sentence names both of your variables and quotes at least one pair of numbers you measured yourself.

**3) Describe** what your graph shows, in words. Name the relationship between your variable and the period, and quote numbers from your own graph. C.ii

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**4) State** whether your points lie close to the line or scatter away from it, and what that tells you about your measurements. C.ii

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### C.iii Was your hypothesis right?

5) **Discuss** whether your data supports the hypothesis you wrote in Part 1. Quote your own numbers — do not just say “yes” or “no”. C.iii

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### C.iv Was your method sound?

#### Required elements for judging your method

Validity is a judgement about **the design and the data**. Tick the box on the side that matches your investigation, then write the sentence underneath.

| Valid if...                                                           | Not valid if...                                                  |
|-----------------------------------------------------------------------|------------------------------------------------------------------|
| <input type="checkbox"/> The points sit close to the line of best fit | <input type="checkbox"/> The points sit a long way from the line |
| <input type="checkbox"/> A clear relationship can be seen             | <input type="checkbox"/> No clear relationship can be seen       |
| <input type="checkbox"/> Repeats were done and an average found       | <input type="checkbox"/> No repeats were done                    |
| <input type="checkbox"/> The instruments worked well throughout       | <input type="checkbox"/> An instrument or a step gave trouble    |
| <input type="checkbox"/> Every control variable stayed the same       | <input type="checkbox"/> Some controls were hard to hold steady  |

My method was \_\_\_\_\_ valid, because \_\_\_\_\_.

**Check.** Your reason points at a row you ticked. Because validity is about the design and the data, the sentence talks about the points, the repeats, the instruments or the controls — not about how much time you had.

6) **Discuss** how much you trust your results. Where could error have crept in, and how big was it compared with the differences you were measuring? C.iv

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7) **State** one thing that was hard to keep the same, and what that could have done to your results. C.iv

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## C.v What would you change?

### Required elements for an improvement

Each improvement gets a reason attached. Four that nearly always apply:

- **Widen the range** — “we tested 5 cm to 15 cm; going to 100 cm would show whether the curve continues”;
- **Take smaller steps** — go up in 5s rather than 10s, so the shape is clearer;
- **Fix whatever was hard to measure accurately** — name the measurement and say what you would do instead;
- **Use a more precise instrument** — one with a smaller resolution, and say what that would let you see.

Then give **one extension**: a new research question with a *different* independent variable, and a sentence on how you would test it.

**Check.** Each improvement has the word *because* or *so* in it, and every one is about the data rather than about the people collecting it.

8) **Describe** two improvements or extensions, and say **why each one would make the investigation better** — an improvement with no reason attached does not score. C.v

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**Before you hand this in.** Read back through Part 2 with the criteria sheet open. For each of C.i to C.v, find the band you think you have reached and *say what would have moved you up one*. That question is the whole reason we started with the rubric instead of finishing with it.